Simple Linear Regression

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Regression Analysis

- Sometimes you want to know the magnitude of the effect of one variable on another variable

- Knowing the rate (magnitude) of change in one variable as a result of the other, means that regression can be used to build a predictive model that can be applied to other situations

- It allows you to predict the value/s of the dependent variable for a given value/s of the independent variable

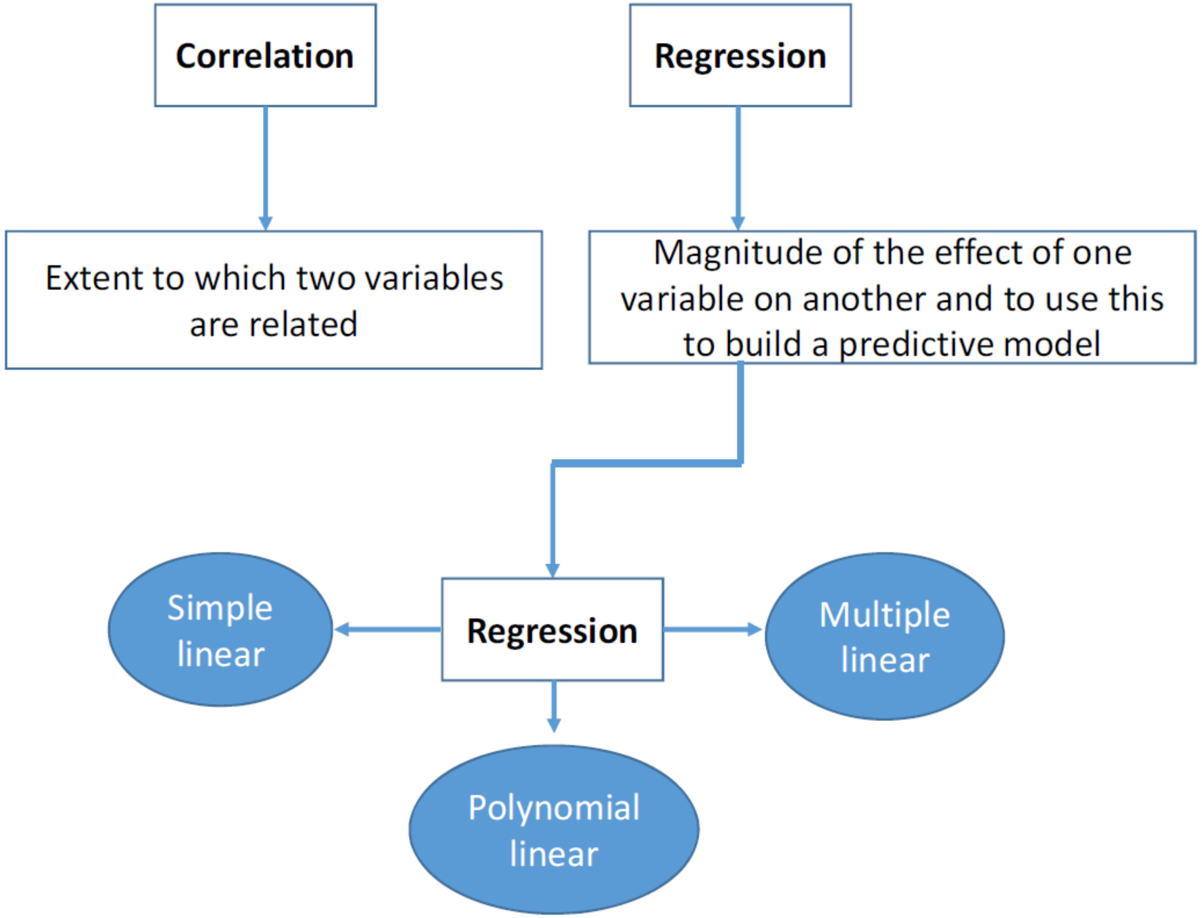
Correlation and regression analysis

Correlation & regression CANNOT be used interchangeably

- The two analyses are based on different assumptions about the nature of the relationship between the two variables examined
- **Regression analysis:** assumes a **cause & effect** relationship. The independent variable is responsible for some of the observed variability in the dependent variable. This cause & effect relationship is assumed to work in only one direction.
  - *E.g., island size can determine species richness of the island, but not vice versa*
- **Correlation analysis:** There is no assumption of cause and effect or dependence of one variable on another (i.e. X does not cause Y to change; X does not affect Y).
- It is only assumed that the two variables are related in some way (i.e. as X changes so does Y but the change in Y is NOT necessarily caused by X).

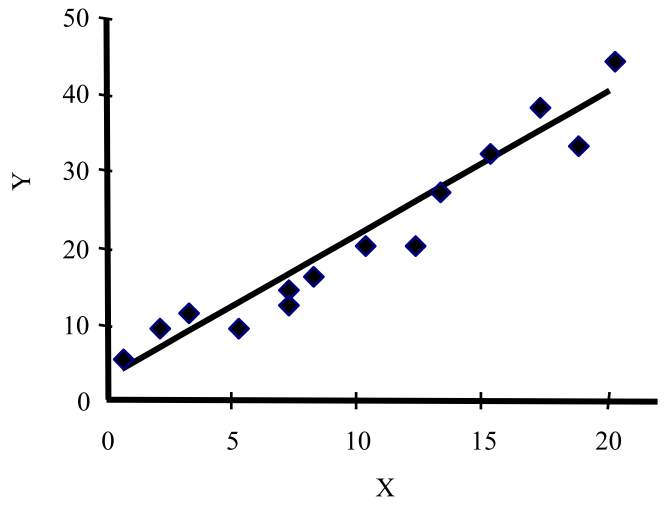
Correlation and regression analysis

Correlation & regression <u>CANNOT</u> be used interchangeably



Simple linear regression

- Simple: 2 variables
- Linear: only linear relationships
- Assumes "cause and effect" relationship therefore:
  - Independent variable or predictor variable (X-axis)
  - Dependent variable or response/outcome variable (Y-axis)

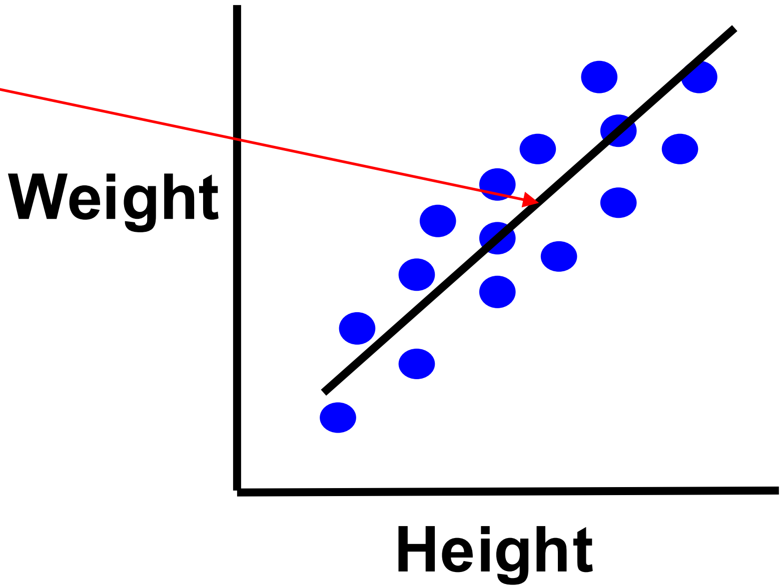


Simple linear regression

Regression analysis fits a 'line of best fit' through the XY data and then analyses the parameters that describe the regression line

- Does the regression model describe the relationship between the data well (is it a good fit for the data)?
- Is the slope significantly different from a zero slope?

Line of best fit through the data = REGRESSION line



Regression Analysis

The regression line is expressed by a linear equation that describes the relationship between X and Y (the regression model)

Two important parameters describe the regression line

Intercept Slope

$$Y_i = \alpha + \beta X_i$$

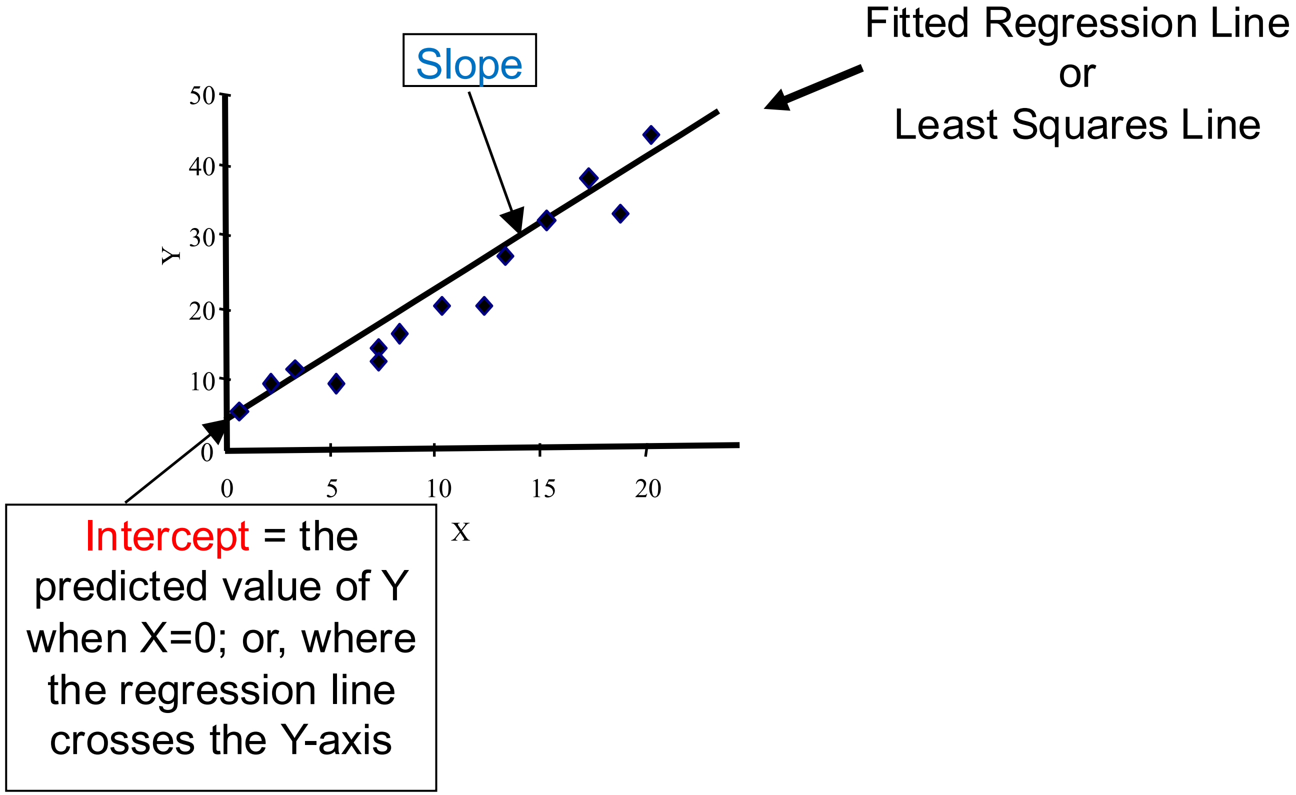
Y_i - i^{th} observation of variable Y (dependent)

X_i - i^{th} observation of variable X (independent)

α - **intercept** of the regression line (the value of

β - the **slope** of the regression line

(change in Y per unit change in X)



Regression Analysis

Two important parameters describe the regression line

Intercept Slope

$$Y_i = \alpha + \beta X_i$$

Slope (β)

The least squares estimate of the slope coefficient (β) of the true regression line is:

$$\beta = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

Shortcut formulas for the numerator and denominator:

$$S_{xy} = \Sigma x_i y_i - (\Sigma x_i)(\Sigma y_i)/n \quad \text{and} \quad S_{xx} = \Sigma x_i^2 - (\Sigma x_i)^2/n$$

Regression Analysis

Two important parameters describe the regression line

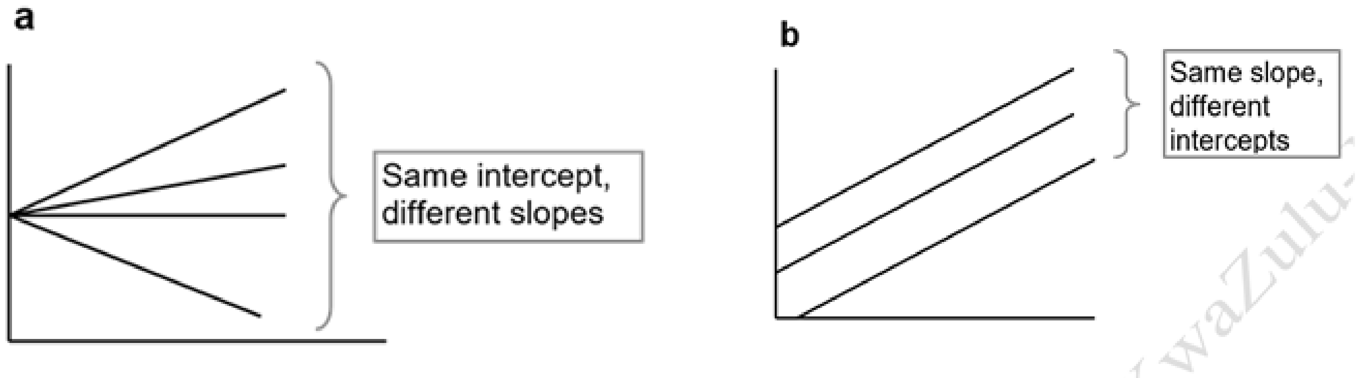
Intercept Slope

$$Y_i = \alpha + \beta X_i$$

Slope (β)

Indicates the **magnitude** of the effect of X on Y i.e., the rate at which Y changes for every 1 unit increase in X

Can be positive, negative, zero (direction of relationship)
If β = 1.86, then Y increases by 1.86 units for every 1 unit increase in X
If β = - 0.92, then Y decreases by 0.92 units for every 1 unit increase in X



Regression Analysis

Two important parameters describe the regression line

Intercept Slope

$$Y_i = \alpha + \beta X_i$$

Intercept (α)

The least squares estimate of the intercept (α) of the true regression line is:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- $\hat{\beta}_0$ **(Estimated Intercept):** The predicted value of y when x is zero. It's where the regression line crosses the vertical axis.
- $\bar{y}$ **(Sample Mean of y):** The average of all observed values for the **dependent variable**.
- $\hat{\beta}_1$ **(Estimated Slope):** The amount y is expected to change for every one-unit increase in x.
- $\bar{x}$ **(Sample Mean of x):** The average of all observed values for the **independent variable**.

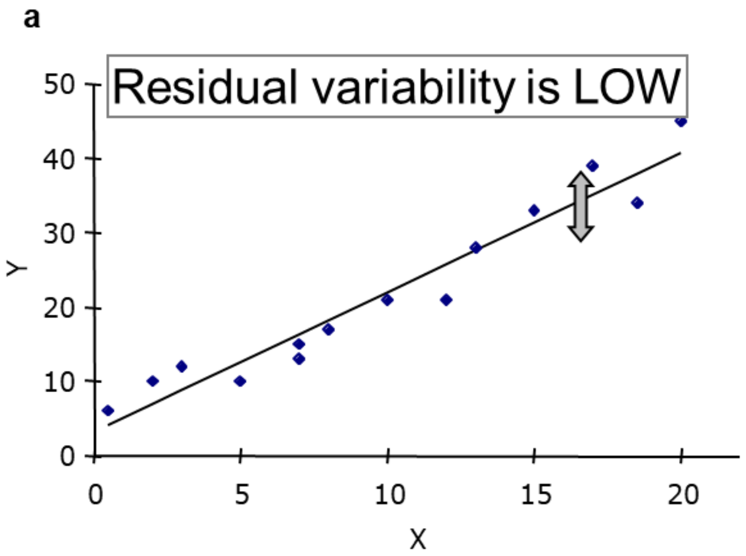
How is a good regression line calculated?

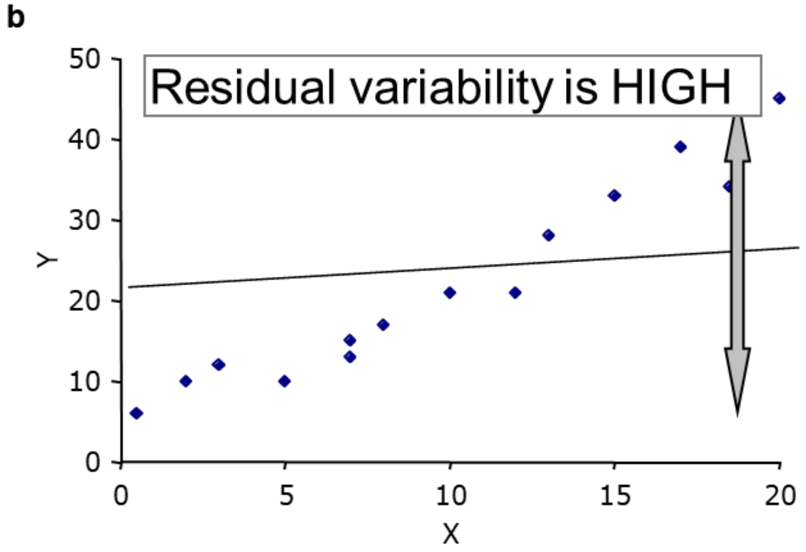
- Most common is Least squares regression

- Goal is to minimize the sum of all the squared deviations between the observed values of Y and the predicted values by the regression line (i.e., it looks for the smallest variability in the residuals around the line)

- The residual is the vertical distance between the observed value (y) and the predicted value ($\hat{y}$), and it is calculated by subtracting $\hat{y}$ from y

- Residuals and least squares regression video: https://www.youtube.com/watch?v=VqD-nf1YUks





How is a good regression line calculated?

Calculating residuals:

- Residual = actual y value – predicted y value

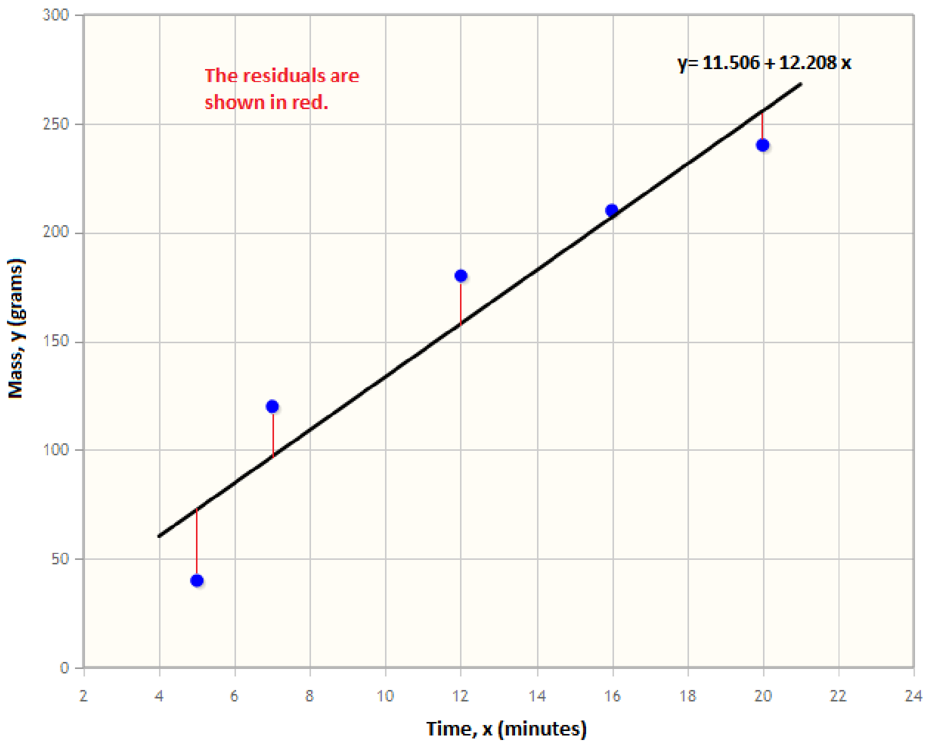
$$r_i = y_i - \hat{y}_i$$

- Regression line equation…

$$\hat{y}_i = a + bx_i$$

- Residual of an observation…

$$r_i = y_i - \hat{y}_i = y_i - (a + bx_i)$$



Calculating a and b…YouTube videos
https://www.youtube.com/watch?v=T7LauV6AnRg
https://www.youtube.com/watch?v=OU4D7FVIWZ8

Video about residuals…
https://www.youtube.com/watch?v=yMgFHbjbAW8

Testing the significance of a regression line

Hypothesis Testing with Regression

Two forms of the general null hypothesis $\mathbf{H_0: \beta = 0}$ can be tested

The slope of the relationship is 0

1. $\mathbf{H_0: \beta = 0}$; X has no effect on Y i.e. *Y* is not dependent on *X*

2. $\mathbf{H_0}$**: *X* and *Y* are not linearly related:** tests appropriateness of whole regression model (intercept and slope) for describing the relationship between *X* & *Y*

Hypothesis Testing with Regression

E.g. Determine whether the number of eggs produced by female fish is affected by water temperature.

- H_A: number of eggs produced is **linearly related** to water temperature
- H_0: number of eggs produced, and water temperature are not linearly related

- H_A: change in water temperature **affects** number of eggs produced
- H_0: change in water temperature has no effect on number of eggs produced

Hypothesis Testing with Regression

Evaluating the slope of the regression

The slope of the regression can be evaluated with *one sample t-test*

1. **H_0: β = 0** is evaluated with a ***t-test***

Compares the slope of the regression line with a hypothetical slope of 0

Measures how many standard errors the estimated sample slope is away from 0

Evaluates significance of the slope (t-statistic)

Slope of the regression line

$$t = \frac{\beta - 0}{s_\beta}$$

Standard error of the slope

Hypothesis Testing with Regression

Evaluating the slope of the regression

Example: If we have a slope: of 1.86 and standard error of the slope 0.16

$$t = \frac{\beta - 0}{s_{\beta}}$$

$$\frac{1.86 - 0}{0.16} = \frac{1.86}{0.16}$$

t = 11.63

df = n - 2

1. Reject the Null Hypothesis (H_0)

- **Condition:** $|t_{\text{calculated}}| \geq t_{\text{critical}}$
- **Action:** You **reject** H_0.
- **Conclusion:** There is a **statistically significant linear relationship** between the independent variable x and the dependent variable y.

2. Fail to Reject the Null Hypothesis (H_0)

- **Condition:** $|t_{\text{calculated}}| < t_{\text{critical}}$
- **Action:** You **fail to reject** H_0.
- **Conclusion:** There is not enough evidence to conclude a linear relationship exists. Changes in x do not reliably predict y in the model.

Hypothesis Testing with Regression

Evaluating linear regression

- Linear regression can be evaluated using an **ANOVA** (Analysis of Variance)
- Evaluates significance of the regression model using the *F-statistic*

2. **H_0: *X* and *Y* are not linearly related** is evaluated using an **ANOVA**

   Calculates the ratio of the variability in the dependent variable that is explained by the regression model to the unexplained variability

Variability in the response variable (*Y*) explained by the regression or due to *X*

$$F = \frac{\text{Regression Variance}}{\text{Residual Variance}}$$

Unexplained variability in the response variable (*Y*) around the regression line

Hypothesis Testing with Regression

Evaluating linear regression

Variability in the response variable (*Y*) explained by the regression or due to *X*

$$F = \frac{\text{Regression Variance}}{\text{Residual Variance}}$$

Unexplained variability in the response variable (*Y*) around the regression line

Total variation = Explained variation + Unexplained variation

Measured by:

Sum of squares total: SST = Sum of squares regression: SSR + Sum of squares error: SSE

$$\sum (Y-\bar{Y})^2 = \sum (\hat{Y}-\bar{Y})^2 + \sum (Y-\hat{Y})^2$$

- Y = Observed value of Y
- $\bar{Y}$ = Mean value of Y
- $\hat{Y}$ = Predicted value of Y (regression line)

Total variation

$$\text{Recall: Variance of Y}: s_Y^2 = \frac{\sum_{i=1}^{n} (Y_i - \bar{Y})^2}{n-1}$$

Degrees of freedom

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

ε_i **(Error Term):** In the real world, data points rarely fall perfectly on a straight line. This term represents the "noise" or the distance between the actual observed data point and the predicted line.

Hypothesis Testing with Regression

Evaluating linear regression

The Setup

Imagine we find that for every hour a student studies, their score goes up by 5 points. Even if they don't study at all, they'd likely get a 40 just by guessing or using prior knowledge.

In this scenario, our equation $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ looks like this:

- y_i **(Exam Score):** The result we want to predict.
- x_i **(Hours Studied):** The input we are measuring.
- $\beta_0 = 40$**:** The "baseline" score if hours studied (x) is zero.
- $\beta_1 = 5$**:** The "boost" per hour of studying.
- ε_i **(The "Human Factor"):** This accounts for things like how much sleep the student got or how lucky they were with the questions.

Hypothesis Testing with Regression

Evaluating linear regression

The Prediction

If a specific student studies for **8 hours**, the formula predicts their score:

$$Score = 40 + (5 \times 8) = 80$$

However, if that student actually scores an **83**, the ε_i **(error term)** for that student is **+3**. The model predicted 80, but the reality was slightly different.

Summary Table

Component	Real-World Meaning
Intercept (β_0)	Starting score with zero effort.
Slope (β_1)	The "payoff" for each hour of work.
Error (ε_i)	Individual luck or outside factors.

Hypothesis Testing with Regression

Evaluating linear regression

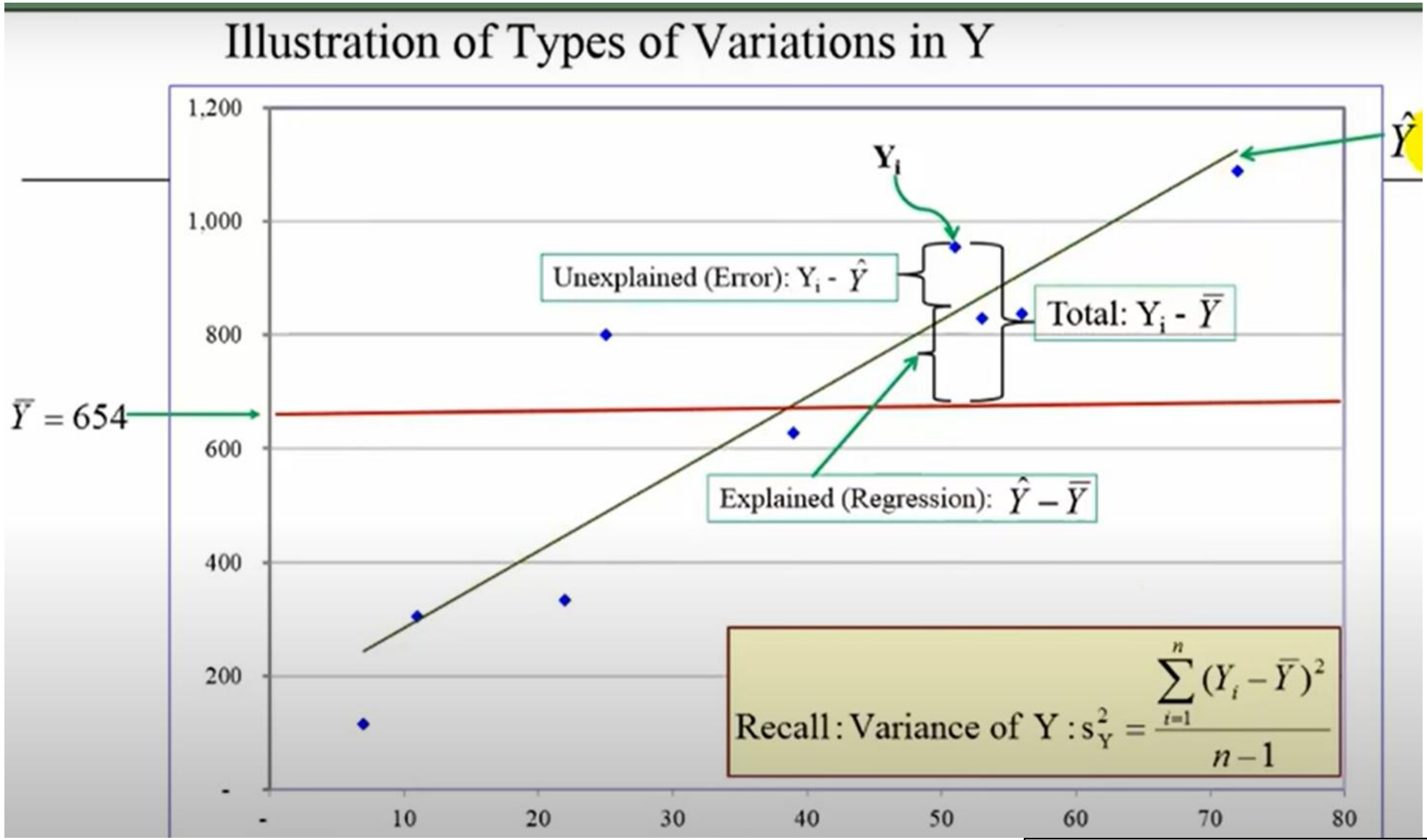
Regression Dataset: Example

	X	Y
	72	1,089
	39	627
	51	954
	25	800
	7	116
	22	334
	11	304
	53	828
	56	838
Total	**336**	**5,890**
Sample mean	**37**	**654**
Sample size: n	**9**	

Source: Pat Obi, Purdue University Northwest

Hypothesis Testing with Regression

Evaluating linear regression



Source: Pat Obi, Purdue University Northwest

One can look up the F-value in an F table just as you can for t-values or F-values in ANOVA

If your F-value is bigger than the one in the table then there is a significant regression slope

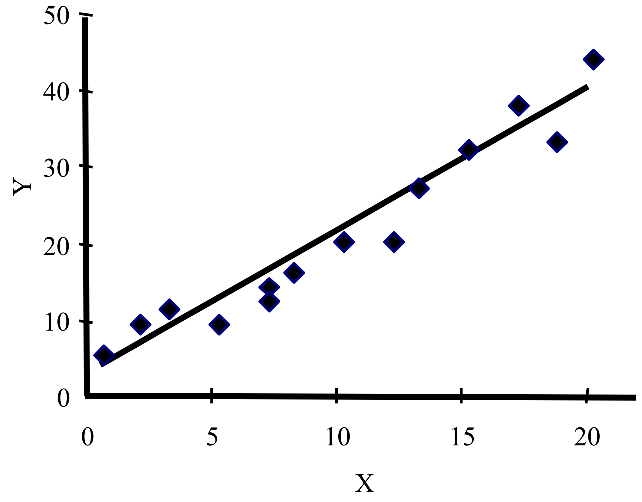
TABLE B.4 (cont.) CRITICAL VALUES OF THE *F* DISTRIBUTION

Numerator DF = 2

α(2): α(1): Denom. DF	0.50 0.25	0.20 0.10	0.10 0.05	0.05 0.025	0.02 0.01	0.01 0.005	0.005 0.0025	0.002 0.001	0.001 0.0005
1	7.50	49.5	200.	800.	5000.	20000.	80000.	500000.	2000000.
2	3.00	9.00	19.0	39.0	99.0	199.	399.	999.	2000.
3	2.28	5.46	9.55	16.0	30.8	49.8	79.9	149.	237.
4	2.00	4.32	6.94	10.6	18.0	26.3	38.0	61.2	87.4
5	1.85	3.78	5.79	8.43	13.3	18.3	25.0	37.1	49.8
6	1.76	3.46	5.14	7.26	10.9	14.5	19.1	27.0	34.8
7	1.70	3.26	4.74	6.54	9.55	12.4	15.9	21.7	27.2
8	1.66	3.11	4.46	6.06	8.65	11.0	13.9	18.5	22.7
9	1.62	3.01	4.26	5.71	8.02	10.1	12.5	16.4	19.9
10	1.60	2.92	4.10	5.46	7.56	9.43	11.6	14.9	17.9
11	1.58	2.86	3.98	5.26	7.21	8.91	10.8	13.8	16.4
12	1.56	2.81	3.89	5.10	6.93	8.51	10.3	13.0	15.3
13	1.55	2.76	3.81	4.97	6.70	8.19	9.84	12.3	14.4
14	1.53	2.73	3.74	4.86	6.51	7.92	9.47	11.8	13.7
15	1.52	2.70	3.68	4.77	6.36	7.70	9.17	11.3	13.2
16	1.51	2.67	3.63	4.69	6.23	7.51	8.92	11.0	12.7
17	1.51	2.64	3.59	4.62	6.11	7.35	8.70	10.7	12.3
18	1.50	2.62	3.55	4.56	6.01	7.21	8.51	10.4	11.9
19	1.49	2.61	3.52	4.51	5.93	7.09	8.35	10.2	11.6
20	1.49	2.59	3.49	4.46	5.85	6.99	8.21	9.95	11.4
21	1.48	2.57	3.47	4.42	5.78	6.89	8.08	9.77	11.2
22	1.48	2.56	3.44	4.38	5.72	6.81	7.96	9.61	11.0
23	1.47	2.55	3.42	4.35	5.66	6.73	7.86	9.47	10.8
24	1.47	2.54	3.40	4.32	5.61	6.66	7.77	9.34	10.6
25	1.47	2.53	3.39	4.29	5.57	6.60	7.69	9.22	10.5
26	1.46	2.52	3.37	4.27	5.53	6.54	7.61	9.12	10.3
27	1.46	2.51	3.35	4.24	5.49	6.49	7.54	9.02	10.2
28	1.46	2.50	3.34	4.22	5.45	6.44	7.48	8.93	10.1
29	1.45	2.50	3.33	4.20	5.42	6.40	7.42	8.85	9.99
30	1.45	2.49	3.32	4.18	5.39	6.35	7.36	8.77	9.90

Regression Analysis

- Measure of Explained Variation ~ The measure of **Goodness of Fit**
- It is the **Coefficient of determination (r^2):** the % of total variation in *Y* that is explained by the regression with *X*
- Gives an indication of the strength of the association and the explanatory power of the model
  - **r^2 = 0.92**
    - 92% of the variability in *Y* can be explained by *X*
    - 8% of the variability in *Y* is unexplained



$$r^2 = \frac{Explained\ Variation}{Total\ Variation} = \frac{SSR}{SST} = \frac{\sum(\hat{Y}-\bar{Y})^2}{\sum(Y-\bar{Y})^2} \Leftrightarrow r^2 = 1 - \frac{SSE}{SST}$$

Regression Analysis

Finding your answers in SPSS output

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.961[a]	.923	.917	3.46541

a. Predictors: (Constant), X

$r^2 = 0.923$; 92% of the variability in Y is explained by X

$$r^2 = \frac{Explained\ Variation}{Total\ Variation} = \frac{SSR}{SST} = \frac{\sum(\hat{Y}-\bar{Y})^2}{\sum(Y-\bar{Y})^2} \Leftrightarrow r^2 = 1 - \frac{SSE}{SST}$$

ANOVA[b]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1735.391	1	1735.391	144.507	.000[a]
	Residual	144.109	12	12.009		
	Total	1879.500	13			

a. Predictors: (Constant), X

b. Dependent Variable: Y

F **= 144.507**; calculated by dividing Mean Square of Regression by Mean Square of Residual (Error)

p=0.000, which should be stated as $p < 0.0005$

There is a significant linear relationship between X and Y

Regression Analysis

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	3.217	1.781		1.807	.096
	X	1.862	.155	.961	12.021	.000

a. Dependent Variable: Y

y-intercept = 3.217
slope = 1.862

t of the slope = 12.021, with $p < 0.0005$

Statistical conclusion: slope of relationship between X and Y is significantly different from 0

Biological conclusion: X has a significant effect on Y; Y increases at a rate of 1.86 for each 1-unit increase in X

$$t = \frac{\beta - 0}{s_{\beta}}$$

Regression Analysis

Predicting values of Y from the regression line

- The equation for the regression line can be used to predict values of *Y*, based on values of X, intercept and slope

- **Example:** In a study comparing species richness with frequency of burning
  - slope (β)= 1.86
  - intercept = 3.22
  - $X_i = 4$
- What is the expected species richness in a plot that was last burned 4 years ago?

$$\hat{y}_i = a + bx_i$$

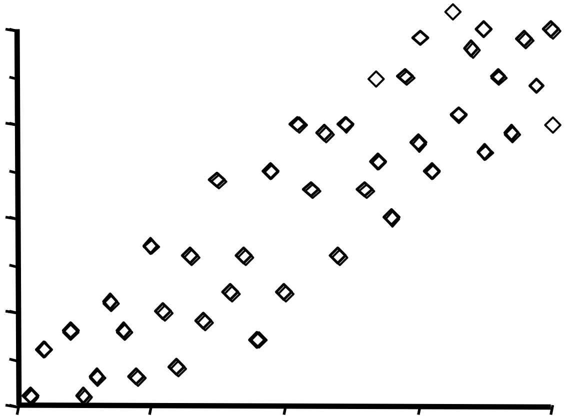
$$Y_i = 3.217 + 1.862\ (4) = 3.217 + 7.44 = 10.66$$

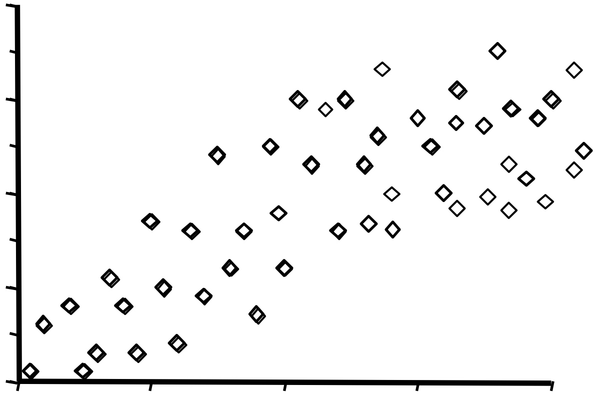
- Should only use when the regression model is **significant,** and you have a **good R^2** – independent variable (X) explains a large proportion of the variability in the dependent (Y)

Assumptions of Linear Regression

1. The relationship between X and Y is linear

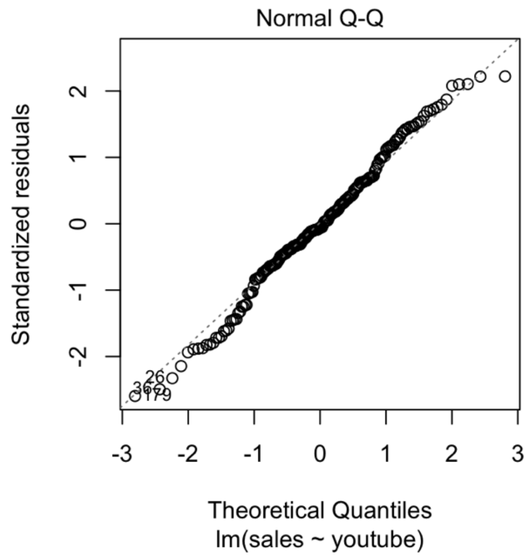
- Check with a scatterplot
- If visually you detect that your data are not linear then transform or
- Use non-linear or polynomial regression analysis





Assumptions of Linear Regression

2. <u>Residuals</u> of the dependent variable are normally distributed
   - Check with a **histogram of the residuals** AND
   - Use Shapiro-Wilk test or Kolmogorov-Smirnov test for normality
   - Shapiro–Wilk test - for small sample sizes (n < 50), but can handle larger sample sizes
   - Kolmogorov–Smirnov test is used for n ≥ 50
   - Or a **Q-Q Plot of the residuals**

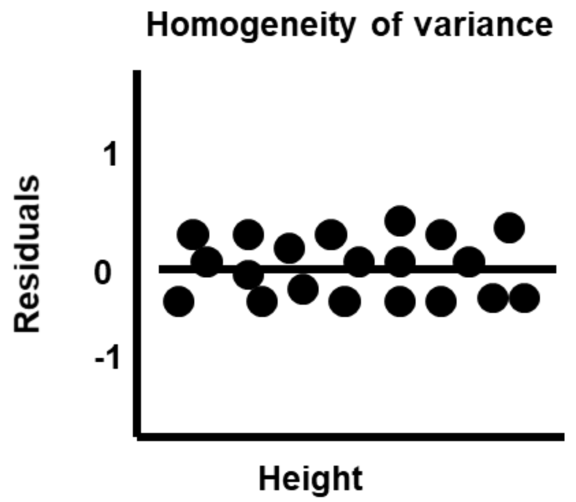


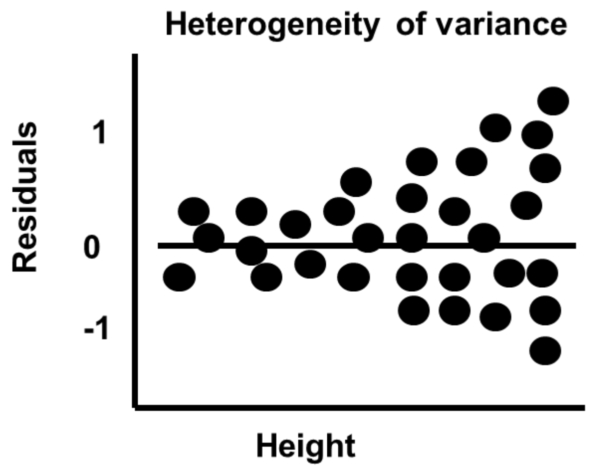
Assumptions of Linear Regression

3. Variance of residuals is equal across range of X

- Use a Levene's test (F-ratio) on the residuals
  - If $p > 0.05$ then assumption of equal variances is satisfied

- Can also plot the residuals in place of the dependent variable
  - If there is homogeneity of variance, then the residuals will be evenly distributed across the x axis and centred on a value of 0





Assumptions of Linear Regression

4. Samples are independent of each other (values in one variable are not influenced by other values in the dataset)

5. No/little measurement error of *X*